



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

Summer Term Examination, AY: 2024-25 (Sem – II)

Class: Second Year B. Tech.	Course: Building Planning and Computer Aided Drawing	Course Code: ESC280
Date: 19.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Instructions to the Students:

- Mobile phones and programmable calculators are strictly prohibited.*
- Figures to the right indicate full marks.*
- Exchange/sharing of stationery, calculator, etc. is not allowed.*
- Assume suitable data wherever necessary.*

Q.N.	Questions	Marks	CO	BT
1.	Write down the functions and role of i) Owner, ii) Civil Engineer, iii) Structural Engineer, and iv) Contractor in construction process.	15	1	1
2.	Explain any Six principles of buildings planning in detail.	15	2	2
3.	Dr. Ram (BAMS) wants to construct a house for him and his family of four members, himself, his wife, 12 years old daughter, and 7 years old son. He is practising at his home where he wants a clinic with waiting area, doctors chamber, panchakarma room, and a pharmacy shop. Along with this he wants a well-planned comfortable house. His plot size is 15 m x 25 m. Draw a line plan.	30	3, 5	6
4.	Draw a line plan for a post office located at a town having population of 40,000.	40	3, 5	5



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination (STE)
Academic Year: 2024-25 (Semester – II)

Class: Second Year

Subject: Theory of Structures - I

Subject Code: PCC-CE-205

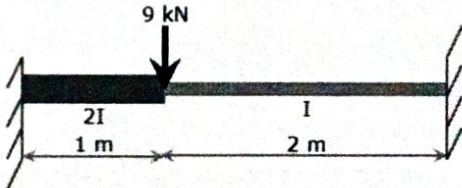
Date: 23-06-2025

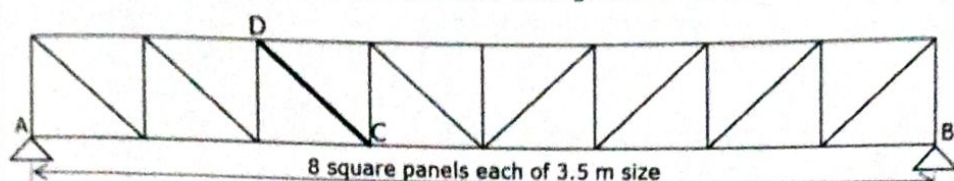
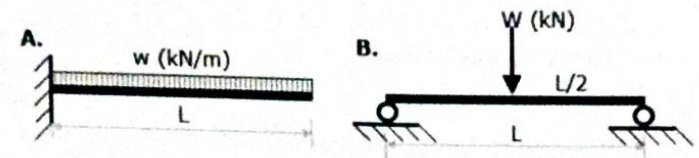
Time: 14:30 to 17:30

Maximum Marks: 100

Note:

- i. Assume appropriate data, if required and state it clearly.
- ii. Non-programmable calculator is allowed.
- iii. Exchange or sharing of any material is not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Question statement	Marks	CO	BT
Q.1	Develop an influence line diagram for maximum bending moment in girder of 20 m at centre and 5 m from left support. Determine maximum bending moment at these points, if a uniformly distributed load of 10 kN/m of 4.0 m length is rolling over that girder.	10	CO3	BT4
Q.2	Develop a neat diagram of the influence line for shear force at a section 3.0 m from one end of a simply supported beam, 12 m long. Use diagram to determine the maximum negative and positive shear force at this section due to uniformly distributed rolling load 5 m long of 2 kN per meter intensity	10	CO3 CO4	BT4
Q.3	A straight pin jointed strut is 50 mm in diameter and 1250 mm long. Determine, a. Euler's crippling load when loaded with point load along the central axis b. The eccentricity which will cause failure at 75% of crippling load, if the yield point of stress of the material is 275 N/mm ² . Assume $E = 2 \times 10^5$ N/mm ² .	10	CO1	BT4
Q.4	A beam of span 3 m is fixed at both the ends and carries a point load of 9 kN at C distant 1 m from support A. The moment of inertia of the portion AC of the beam is 2I and that of CB is I. Determine the fixed end moment and reactions. Use minimum strain energy concept. 	10	CO2	BT4
Q.5	ABCD is a continuous beam of 18 m is divided equally in three spans. Each span is loaded centrally with a concentrated load of 50 kN. Additionally, entire beam is loaded with a uniformly distributed load of 20 kN/m. Determine the reaction and moment at support in beam using Clapeyron's theorem.	10	CO2 CO3	BT4
Q.6	A suspension cable 140 m span and 14 m central dip carries a load of 1 kN/m. Determine the maximum and minimum tension in the cable. Determine the horizontal and vertical forces in each pier under following conditions. 1. If the cable passes over a frictionless roller on the top of the piers 2. If the cable is firmly clamped to saddles carried on frictionless rollers on the top of the piers. In both case the back stay is inclined at 30° with the horizontal.	10	CO2	BT4

Q.N.	Question statement	Marks	CO	BT
Q.7	A pratt girder shown in figure below consist of eight panels of each 3.5 m square, the loading being on the lower boom. Develop an influence line diagram for the force in the member CD. Determine the maximum tension and maximum compression in CD due to a concentrated rolling load of 20 kN. 	10	CO4	BT4
Q.8	Develop conjugate beam for following beams. 	10	CO1	BT3
Q.9	Determine the deflection of beam under the concentrated load. The beam is simply supported with point load of 90 kN at point C. The beam AB has length of 4.5 m, and point C is 2.7 m away from the support A. (Use conjugate beam method) (Assume $E = 210 \times 10^6 \text{ kN/m}^2$ and I of 160×10^{-6} and $120 \times 10^{-6} \text{ m}^4$ for section AC and CB, respectively.)	10	CO1	BT4
Q.10	Using strain energy method determine the deflection at centre of fixed beam of length 'L'. Consider the entire beam is loaded with uniformly distributed load of 'w'.	10	CO1	BT4

***** ALL THE BEST *****

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

COs	On successful completion of this course, students should be able to:
CO1	Recall the concept of deflection, bending moment and shear force diagram in beams, frames, trusses, and columns under various loading conditions using different analysis methods.
CO2	Express knowledge to determine forces in determinate and indeterminate structures
CO3	Apply different analysis methods to obtain shear force, bending moment.
CO4	Calculate different stress variants for the given loading condition on beams, frames, trusses, columns and to perform ILD analysis of determinate beams and trusses.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY

Course: Fluid Mechanics

Course Code: PCC-CE 203

Date: 20.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

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Q.N.			Marks	CO	BT
Q.1	a	Define (i) Mass density; (ii) Weight density; (iii) Specific volume; (iv) Viscosity; (v) Capillarity; (vi) Vapour Pressure	06	CO1	BT1
	b	Derive the expression for capillary rise of liquid.	05	CO1	BT3
	c	A 15cm diameter vertical cylinder rotates concentrically inside another cylinder of diameter 15.1 cm. Both cylinders are 25 cm high. The space between the cylinders is filled with a liquid whose viscosity is unknown. If a torque of 12 Nm is required to rotate the inner cylinder at 100 r.p.m., determine the viscosity of the fluid.	06	CO1	BT3
Q.2	a	State and derive Pascal's law.	06	CO2	BT3
	b	Define (i) Absolute pressure; (ii) Gauge pressure; (iii) Atmospheric pressure; (iv) Vacuum pressure	04	CO2	BT2
	c	A body of dimensions 1.5 m* 1.0 m*2 m, weighs 1962 N in water. Find its weight in air. What will be its specific gravity?	06	CO2	BT3
Q.3	a	A 25 cm diameter pipe carries oil of sp.gr.0.9 at a velocity of 3 m/s. At another section the diameter is 20 cm. Find the velocity at this section and also mass rate of flow of oil.	06	CO3	BT3
	b	Define (i) Rotational and Irrotational flow; (ii) Compressible and Incompressible flow; (iii) Steady and Unsteady flow	06	CO3	BT2
	c	What is flow net? Explain the relation between stream function and velocity potential function.	05	CO3	BT2
Q.4	a	Derive Euler's equation of motion along a stream line for an ideal fluid stating clearly assumptions.	08	CO3	BT3
	b	A pipe through which water is flowing, is having diameters 20 cm and 10 cm at the cross section 1 and 2 respectively. The velocity of water at section 1 is given 4 m/s. Find the velocity head at sections 1 and 2 and also rate of discharge.	04	CO3	BT3
	c	What is vortex flow? Explain forced vortex and free vortex flow	05	CO1	BT2
Q.5	a	Describe the following terms: (i) Coefficient of velocity (C_v) (ii) Coefficient of contraction (C_c) (iii) Coefficient of discharge (C_d)	06	CO1	BT2
	b	A rectangular orifice 0.9 m wide and 1.2 m deep is discharging water from a vessel. The top edge of the orifice is 0.6 m below	06	CO4	BT3

		the water surface in the vessel. Calculate the discharge through the orifice if $C_d=0.6$ and percentage error if the orifice is treated as a small orifice.			
	c	Define orifice. Classify the orifice based on size, cross section, nature of discharge and shape of upstream edge	05	CO4	BT2
Q.6	a	An oil of sp.gr.0.7 is flowing through a pipe of diameter 300 mm at the rate of 500 litres/s. Find the head lost due to friction and power required to maintain the flow for a length of 1000 m. Take $\nu=0.29$ stokes.	08	CO5	BT3
	b	Explain the concept of Hydraulic Gradient Line and Total Energy Line	04	CO1	BT2
	c	Three pipes of lengths 800m, 500 m and 400 m and of diameters 500 mm, 400 mm and 300 mm respectively are connected in series. These pipes are to be replaced by a single pipe of length 1700 m. Find the diameter of the single pipe.	04	CO5	BT3

==== ALL THE BEST ====

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Define and understand the fundamental concepts of fluid mechanics
CO 2	Analyse the hydrostatic forces and stability of floating bodies.
CO 3	Apply Bernoulli's equation to fluid flow problems.
CO 4	Use of different fluid flow measuring devices
CO 5	Apply appropriate equations and principles to analyse pipe flow problems



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY

Course: Fluid Mechanics

Course Code: PCC-CE 202

Date: 20.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

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Q.N.			Marks	CO	BT
Q.1	a	Define (i) Mass density; (ii) Weight density; (iii) Specific volume; (iv) Viscosity; (v) Capillarity; (vi) Vapour Pressure	06	CO1	BT1
	b	Derive the expression for capillary rise of liquid.	05	CO1	BT3
	c	A 15cm diameter vertical cylinder rotates concentrically inside another cylinder of diameter 15.1 cm. Both cylinders are 25 cm high. The space between the cylinders is filled with a liquid whose viscosity is unknown. If a torque of 12 Nm is required to rotate the inner cylinder at 100 r.p.m., determine the viscosity of the fluid.	06	CO1	BT3
Q.2	a	State and derive Pascal's law.	06	CO2	BT3
	b	Define (i) Absolute pressure; (ii) Gauge pressure; (iii) Atmospheric pressure; (iv) Vacuum pressure	04	CO2	BT2
	c	A body of dimensions 1.5 m* 1.0 m*2 m, weighs 1962 N in water. Find its weight in air. What will be its specific gravity?	06	CO2	BT3
Q.3	a	A 25 cm diameter pipe carries oil of sp.gr.0.9 at a velocity of 3 m/s. At another section the diameter is 20 cm. Find the velocity at this section and also mass rate of flow of oil.	06	CO3	BT3
	b	Define (i) Rotational and Irrotational flow; (ii) Compressible and Incompressible flow; (iii) Steady and Unsteady flow	06	CO3	BT2
	c	What is flow net? Explain the relation between stream function and velocity potential function.	05	CO3	BT2
Q.4	a	Derive Euler's equation of motion along a stream line for an ideal fluid stating clearly assumptions.	08	CO3	BT3
	b	A pipe through which water is flowing, is having diameters 20 cm and 10 cm at the cross section 1 and 2 respectively. The velocity of water at section 1 is given 4 m/s. Find the velocity head at sections 1 and 2 and also rate of discharge.	04	CO3	BT3
	c	What is vortex flow? Explain forced vortex and free vortex flow	05	CO1	BT2
Q.5	a	Describe the following terms: (i) Coefficient of velocity (C_v) (ii) Coefficient of contraction (C_c) (iii) Coefficient of discharge (C_d)	06	CO1	BT2
	b	A rectangular orifice 0.9 m wide and 1.2 m deep is discharging water from a vessel. The top edge of the orifice is 0.6 m below	06	CO4	BT3

		the water surface in the vessel. Calculate the discharge through the orifice if $C_d=0.6$ and percentage error if the orifice is treated as a small orifice.			
	c	Define orifice. Classify the orifice based on size, cross section, nature of discharge and shape of upstream edge	05	CO4	BT2
Q.6	a	An oil of sp.gr.0.7 is flowing through a pipe of diameter 300 mm at the rate of 500 litres/s. Find the head lost due to friction and power required to maintain the flow for a length of 1000 m. Take $\nu=0.29$ stokes.	08	CO5	BT3
	b	Explain the concept of Hydraulic Gradient Line and Total Energy Line	04	CO1	BT2
	c	Three pipes of lengths 800m, 500 m and 400 m and of diameters 500 mm, 400 mm and 300 mm respectively are connected in series. These pipes are to be replaced by a single pipe of length 1700 m. Find the diameter of the single pipe.	04	CO5	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Define and understand the fundamental concepts of fluid mechanics
CO 2	Analyse the hydrostatic forces and stability of floating bodies.
CO 3	Apply Bernoulli's equation to fluid flow problems.
CO 4	Use of different fluid flow measuring devices
CO 5	Apply appropriate equations and principles to analyse pipe flow problems



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY

Course: Hydraulic Engineering

Course Code: PCC-CE 202

Date: 21.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
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Q.N.			Marks	CO	BT
Q.1	a	Define following terms: (i) Sub-critical, Critical and Super Critical flow (ii) Non-uniform flow, Rapidly varied flow, Gradually varied flow	05	CO1	BT1
	b	A trapezoidal channel has side slopes of 3 horizontal to 4 vertical and slope of its bed is 1 in 2000. Determine the optimum dimensions of the channel, if it is to carry water at $0.5 \text{ m}^3/\text{s}$. Take Chezy's constant as 80.	06	CO1	BT3
	c	Derive the expression for loss of energy due to hydraulic jump.	06	CO1	BT3
Q.2	a	What do you mean by Prandtl's maxing length theory?	05	CO2	BT2
	b	Explain the hydrodynamically smooth and rough boundaries.	04	CO2	BT2
	c	Describe the step by step procedure of Hardy Cross method for pipe network problems.	08	CO2	BT2
Q.3	a	Define: (i) Boundary Layer Thickness; (ii) Displacement Thickness; (iii) Momentum Thickness; (iv) Energy Thickness	08	CO3	BT2
	b	Find the momentum thickness and energy thickness for the velocity distribution in the boundary layer given by $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity at a distance y from the plate and $u = U$ at $y = \delta$, where δ = boundary layer thickness. Also calculate the value of $\frac{\theta}{\delta^{**}}$	08	CO3	BT3
Q.4	a	The pressure difference Δp in a pipe of diameter D and length l due to viscous flow depends on the velocity V , viscosity μ and density ρ . Using Buckingham's π -theorem, obtain an expression for Δp .	08	CO4	BT3
	b	What is similarity laws? Explain Reynold's model law.	04	CO4	BT2
	c	Define : (i) Mach number; (ii) Euler's number; (iii) Weber's number; (iv) Froude's number	04	CO4	BT2
Q.5	a	Derive the expression of work done by the centrifugal pump on water	06	CO5	BT3
	b	Define : (i) Manometric Efficiency (ii) Mechanical efficiency (iii) Overall efficiency	03	CO5	BT2
	c	Two geometrically similar pumps are running at the same speed of 1000 r.p.m. One pump has an impeller diameter of 0.30 meter	08	CO5	BT3

		and lifts water at the rate of 20 litres per second against a head of 15 meters. Determine the head and impeller diameter of the other pump to deliver half the discharge.			
Q.6	a	Draw the neat diagram of reciprocating pumps and list the all components of it.	05	CO5	BT2
	b	What is submersible pump? Write the working principle of submersible pump	04	CO5	BT2
	c	A double acting reciprocating pump, running at 40 r.p.m. is discharging 1 m ³ of water per minute. The pump has stroke of 400 mm. The diameter of piston is 200 mm. The delivery and suction heads are 20 m and 5 m respectively. Find the slip of the pump and power required to drive the pump.	08	CO5	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply the knowledge of fluid mechanics to address the uniform flow problems in open channels
CO 2	Solve flow problems of turbulent flow, pipe network analysis and water hammer
CO 3	Recognize concept of laminar and turbulent boundary layer and its analysis
CO 4	Perform dimensional and model analysis for different fluid flow problems
CO 5	Understand the functions and working principles of hydraulic machineries

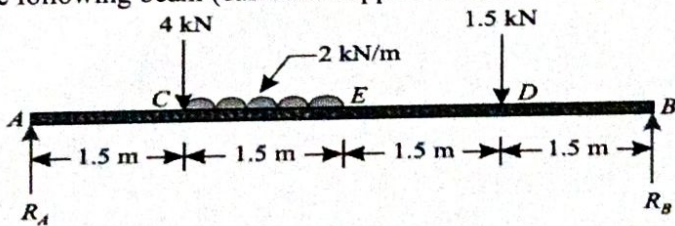


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer term Examination
Academic Year: 2024-25

Class: SY	Subject: Strength of Materials	Subject Code: PCC-CE201
Date: 24.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Mobile phones and programmable calculators are strictly prohibited.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.
- v. All questions are compulsory

Q.N.		Marks	CO	BT
Q.1	a Explain the following terms 1) Moment of resistance 2) Neutral axis 3) Nature of distribution of bending stress 4) Hoop stress 5) Principal plane	15	CO2, CO3	BT1
Q.2	a Write short note on 1) Mohr's circle 2) Distribution of shear stress on a rectangular section 3) Combined effect of bending and torsion	15	CO3, CO5	BT2
Q.3	a While testing on metallic rod it is observed that diameter of the rod reduced by 0.0030 mm under an axial pull of 30 kN. The original diameter of the rod is 15 mm. If rigidity modulus of the rod metal be 70 kN/mm ² . Find young's modulus and bulk modulus.	10	CO1	BT4
Q.4	a Draw Shear force diagram (BMD) and Bending moment diagram of the following beam (calculate support reactions first) 	10	CO5	BT5
Q.5	a A solid steel shaft has to transmit 80 KW at 350 rpm. Taking allowable shear stress 75 MN/m ² . Find suitable diameter for the shaft, if the maximum torque is equal to the average torque.	10	CO3	BT4
Q.6	a A thin cylindrical shell is to be made of 25 mm thick plate having tensile stress of 150 MN/mm ² . Calculate permissible diameter of the shell for internal pressure of 2MN/m ² .	10	CO2	BT4

Q.7	a	A column 18 m long has a circular section of 60 cm diameter. The column is made of a metal having modulus of elasticity ($E = 2 \times 10^4 \text{ KN/cm}^2$). Use Euler's formula to determine the critical load if 1) Both ends of column fixed 2) Both ends of column hinged 3) One end fixed another end free 4) One end fixed another end hinged	10	CO4	BT3
Q.8	a	At a point on a material body the stresses are $\sigma_x = (-60 \text{ Mpa})$, $\sigma_y = 120 \text{ Mpa}$ and $\tau_{xy} = 40 \text{ Mpa}$. Draw the stress diagram of the material body and calculate 1) Stresses acting on an element inclined at an angle $\theta = 30^\circ$ 2) The principal stresses and 3) The maximum shear stress	10	CO4	BT4
Q.9	a	A steel plate is bent into a circular arc of radius 15 m. If the plate section be 100 mm wide and 20 mm deep, ($E = 2 \times 10^5 \text{ N/mm}^2$) find i) maximum bending stress induced ii) the bending moment which can produce iii) loading (W KN) if beam is cantilever having 6 m length and point load at the free end Draw bending stress distribution along the depth of the beam	10	CO3	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall stress and strain fundamentals, Hooke's law, principal stresses, strain energy, and column buckling.
CO 2	Understand how load intensity, stress, strain, bending moment, and shear force influence structural member behaviour under different loads.
CO 3	Apply pure bending and torsion principles to calculate bending and shear stresses in beams and shafts.
CO 4	Analyse stress distribution in beam cross-sections, principal stresses and determine critical loads for column buckling using Euler's and Rankine's formulae.
CO 5	Evaluate and interpret shear force and bending moment diagrams for complex beam configurations and evaluate the combined effect of bending and torsion.